Dynamic Systems Theory: An Improved Model of Cognitive Processing

Emily Tieu

Department of Cognitive Science, University of Toronto

COG250: Introduction to Cognitive Science

April 7, 2025

Abstract

Dynamical Systems Theory (DST) is a model of human cognitive processing via a system of nonlinear differential equations. Cognition is best detailed by the patterns mapped by these systems, allowing both a quantitative and qualitative understanding of real-time processing. Points on a DST model are plotted on a state space, a system of internal and external inputs over time that affect thoughts and behaviours. These points follow a trajectory, culminating at an attractor that is most influential to processing. On a model of neurotic behaviour, these attractors might include conflicting stressors and coping mechanisms that each cause variability in the resulting system. Moreover, applications of DST can be used for analyzing physical and mental development over time. In contrast to previous models of development, DST enables a continuous shift between stages of development, accounting for individual differences. Dynamic systems can also be examined for equilibrium patterns. As fluctuations in cognitive processes and the environment are imminent, it is important for humans to balance stability and flexibility mechanisms to appropriately adapt to changes they encounter. Consequently, DST entails inclusivity in individual differences, making it a better alternative to discrete models of cognition.

Dynamic Systems Theory: An Improved Model of Cognitive Processing

Dating back to the earliest of philosophers, the world has been theorized to be composed of both material and mental substances that are embodied by physical extensions and inner thoughts. Through individual lenses, such substances interact with each other by way of cognition, a real-time, continuous process that could not otherwise be replicated by discrete models. Cognitive function occurs simultaneously and non-linearly, making it difficult to detail the fluctuations in our thoughts and emotions without a continuous mapping. Most prevalent to cognition is the existence of biological systems, which interact with and are affected by the influence of the environment (Favela, 2020). Consequently, our cognitive processes cannot be understood simply using a reductionist approach, a means of reducing a subject into its constituent parts (Favela, 2020). Instead, each component interacts with each other to form an emergent system with varying influences that evolve over time (McClelland, et al., 2009; Samuelson, et al., 2015). By integrating time and systems components, Dynamical Systems Theory (DST) corroborates aspects of mathematics and cognitive science to scrutinize the change of abstract and physical systems over time (Favela, 2020). DST employs different variables over time in the form of differential equations to study the patterns and evolution of a system over time (Favela, 2020). Thus, these models are capable of exhibiting long-term and short-term behaviours, stability, and states of processing (McClelland, et al., 2009).

Furthermore, Dynamic Systems Theory best describes the human ability to self-organize as a result of emergent processes to attain homeostasis (Keenan, 2010). At its core, self-organization is invoked by the activation and connection of neurons through an internal or external stimulus (Keenan, 2010). These individual processes conglomerate on a much larger scale to form perceptions, most notably affecting one's stability and changes in both internal and external systems (Keenan, 2010). Thus, cognition revolves around how these different stimuli interact with each other, often with varying strengths, in order to induce different states. In DST, these states only exist momentarily as there is constantly a change in the systems (Perone, et al., 2021). In fact, Perone, et al. (2021) describe systems to be historical, where repeated entry into a similar state will cause that particular state to become preferred and therefore engrained in one's development. Such preferences give rise to the idea of attractors, which is critical to the stability of emergent behaviours and self-organization (McClelland, et al., 2009; Keenan, 2010). This leads to a circular causality effect of the brain's subsystems simultaneously influencing and coordinating with one another as both top-down and bottom-up processes (Keenan, 2010; Perone, et al., 2021). This allows for flexible adaptations of social-emotional traits, hence encouraging the development of personality traits and responses to stressful environments (Perone, et al., 2021; Keenan, 2010).

While it is evident that Dynamic Systems Theory can model real-time cognitive processes, it remains unclear how well DST contributes to understanding human functioning under varying mental states, thus begging the question: To what extent does the efficacy of cognitive processing affect overall functioning as described by dynamic systems? In response to levels of stress, DST best depicts changes in cognitive processing at a definitive and non-arbitrary scale, overtaking its contending models. Moreover, DST prevails in describing development, where analyzing dynamic systems allows for a better understanding of stable and unstable conditions that cannot be replicated via discrete models.

Stress Responses

The human body is inherently designed to experience and react to instances of stress, whether internal or external (Li, et al., 2024). However, the continued activation of stress will cause the body to exert a surplus of hormones to increase tension and alertness, leading to exhaustion, a decline in personal resources, and regressions in functioning (Li, et al., 2024; Keenan, 2010). With the randomness of this cause-and-effect relationship, Li, et al. (2024) assert that the human stress response is a complex system of nonlinear chaos. Therefore, methods of nonlinear dynamic systems are better appropriate to delineate changes in stress as opposed to deterministic linear methods (Li, et al., 2024).

Keenan (2010) explicates that Dynamic Systems Theory places an emphasis on the relationship between stress and coping mechanisms among internal processes. According to Keenan (2010), each individual maintains a storage of personal resources which hold crucial value on how they cope with stress. These resources vary significantly from person to person, taking on forms such as initiative, avoidance, or even humour (Keenan, 2010; Benight, et al., 2017). To explain such coping mechanisms, Benight, et al. (2017) discuss self-evaluation techniques as a key to predicting behaviour across several domains of cognitive processing. Self-evaluation unveils self-efficacy perception, which is essential to overall motivation. Hence, self-efficacy plays a decisive role in perseverance and constructive goal-setting (Benight, et al., 2017). Additionally, Keenan (2010) expounds on this notion by stating that the agglomeration of parts that comprise self-efficacy and appraisal is stabilized into a mental state upon joining with an emotion. That is, positive interpretations will generate personal resources and coping pathways to respond to stress, ultimately culminating in a sense of hope (Keenan, 2010). DST also models goals and intentions as attractors in a system, exhibiting that our values are one of the most influential elements in perceiving and dealing with stress (Keenan, 2010). It can therefore be deduced that we tend to cope towards our values, a nonlinear variable that fluctuates over time.

Dynamic systems use differential equations to illustrate sudden shifts in the organization of one's cognitive processes. With a set of stress variables, dynamic systems can model resulting behaviours at any given time, where future configurations can be determined by previous configurations (Favela, 2020). These variables are considered to be continuous functions themselves, allowing each variable to have their own nonlinear influence in the overarching system (Favela, 2020). More recently, Chaos Theory has emerged as an extension of nonlinear dynamic systems. Chaotic systems are special in that they possess a strong sensitivity to variations in a system's initial conditions, where a small change may cause a spike in the resulting system (Li, et al., 2024). Thus, Dynamic Systems Theory recognizes that although chaos systems may be unpredictable in the long run, they can provide insight into local trajectories of a state space and what variables might reconstruct a system's attractors (Favela, 2020; Li, et al., 2024). Li, et al. (2024) identify a source of chaos in stress systems to stem from positive and negative events. Intensity of an event and proximity to deadlines were both found to induce stress, as well as different coping behaviours, including sleep duration and mood. However, stress intensity and behaviour is also dependent upon individual personality traits, where neuroticism and extraversion can play a significant role (Li, et al., 2024). Hence, chaotic systems are not all random, as these internal laws determine proceeding actions, allowing short-term stress to be predictable. It is thus apparent that DST is able to quantify the vulnerability, complexity, and unpredictability of a system as complex as the brain's mental processes.

By utilizing DST and its corresponding nonlinear differential systems, we acquire a much more accurate account of how different components of stress and coping affect our mental states. As opposed to linear difference equations, which only serve as an approximation, DST perceives the variability in self-regulation as the capacity to return to normal functioning by recognizing falsifications and nonlinear shifts (Favela, 2020; Benight, et al., 2017). With stress being a chaotic system in itself, Dynamic Systems Theory prevails over discrete systems in modelling competing behavioural tendencies.

Developmental Changes

Dynamic Systems Theory demonstrates that development is nonlinear and an endurance of distress, fluidity, and disequilibrium (Knight, 2014). Knight (2014) notes that with the internal changes coupled with adolescence comes a general fragility of the ego, causing possible maladaptive coping behaviours. While some character traits and temperament are inherent, they too are vulnerable to continual change as a child responds to their changing environment (Knight, 2014). Consequently, Knight (2014) proposes that we reject developmental theories that progress sequentially through discrete stages, arguing that development, by no means, follows a linear timeline. Instead, using dynamic systems better entails the nonlinearity of experiences, where certain events might cascade into larger consequences for some while having no effect on others (Knight, 2014). Keenan (2010) further states that the DST scaffold delineates how children shift from one stage to another, complementing the developmental theories' mere explanation of the stages. As such, development is no longer limited by predetermined phases of what is considered normal, allowing infinitely many ways for individuals to respond to internal and external changes (Knight, 2014).

Moreover, applications of DST have transformed our understanding of motor development and object permanence (Perone, et al., 2021; Samuelson, et al., 2015). To demonstrate this claim, Perone, et al. (2021) illustrate several examples of the development of children's goal-oriented behaviour under a simple sorting task. One case discovered that providing children with prior exposure to the colours and shapes used in the sorting task allowed them to more feasibly complete the sorting (Perone, et al., 2021). As experience over these dimensions strengthens neural connectivity, there exists a new attractor state that uses the memory traces accumulated with the prior experience (Perone, et al., 2021). Similarly, Perone, et al. (2021) noted that children's linguistic environment in their earliest years of life provides more exposure to colour than shape. As a result, children did significantly worse at sorting tasks labelled with shapes. A DST analysis of this phenomenon observes that the children's history strengthened their connections between colour and object representation before developing connections to shape (Perone, et al., 2021). Experience, as contextualized by DST, thus has a profound effect on development and goal-oriented behaviours.

Moreover, Samuelson, et al. (2015) scrutinize the development of object permanence using the A-not-B error, an occurrence in which infants will continue searching for an object in its previous location despite seeing it elsewhere. Object representation using previous approaches is suggested to be a behavioural change of cognitive processes; infants 10 months of age lacked the understanding of the object's continued existence after it was hidden, while infants 12 months of age were able to conceptualize it (Samuelson, et al., 2015). However, Dynamic Systems Theory implies that behavioural change is also influenced by varying changes in motor development (Samuelson, et al., 2015). Motor control was also found to drive goal-directed behaviour of locating the object, where changes in motor skills during the experiment and over the course of development both affected the assembly of systems involved in object representation (Perone, et al., 2021; Samuelson, et al., 2015). Furthermore, the multidimensionality of dynamic systems involving both mental and motor skills allows for the development of discipline in children. For instance, sitting still requires the simultaneous assembly of muscle activity, emotional regulation, motivation, and other cognitive processes to dictate behaviour (Perone, et al., 2021). Therefore, DST reinforces that development is a multidimensional agglomeration of mental and bodily dynamics.

As most developmental theories fail to explain the shifts that occur between stages, DST fills in these novel gaps. By accounting for history, individual internal and external processes, and other stressors, a more complex understanding of development is attained. Consequently, it is identified that changes to the development of one component can cause changes to the organization of the whole system (Perone, et al., 2021). As opposed to discrete models, Dynamic Systems Theory provides a plethora of trajectories to follow in response to any given event, allowing individual differences to stand out.

Stability vs. Flexibility

With the immense interconnectivity of the brain's cognitive processes, its resistance to perturbations is critical. As a dynamic system, the sensitivity of our functioning systems to chaos stressors necessitates a form of protection against these distractors (Schoner, 2009). Under the influence of a complex environment and an equally complex nervous system, humans have more degrees of freedom than needed for any singular task (Schoner, 2009). Consequently, stability is crucial to successful task performance and behavioural adaptation (Braun, et al., 2015). Mathematically, perturbations drive dynamical systems away from an attractor that they would otherwise converge to (Schoner, 2009). To restore balance, Schoner (2009) discusses the existence of stabilization mechanisms to push against perturbation forces. Intrinsically, neural networks have natural mechanisms to stabilize attractors (Schoner, 2009). On the other hand, Smith (2009) argues that too much stability may also cause inflexibility, disabling a response to a change in sensory input. In fact, stability may even lead to a deficiency in children's cognition as they rely too heavily on past experiences (Smith, 2009). To find a balance between stability and flexibility, an extension of the Dynamic Systems Theory can be employed to better understand how a system reaches equilibrium and bifurcation states.

As the Dynamic Systems Theory seeks to examine complex patterns of change over time, it is thus applicable to studying patterns of temporal stability associated with neuroticism. In using a DST approach to analyze the cognitive model of suicide, Rogers, et al. (2024) ascertained that feelings of hopelessness can destabilize trajectories of stress, suicide rumination, and suicidal intent. Additionally, when hopelessness and stress are elevated, voluntarily processing suicide-related thoughts can increase negative thinking and overtake concurrent information processing (Rogers, et al., 2024). While there is a need for internal or external intervention to restabilize the system, DST provides a basis for catalysts of cognitive change. It is prevalent that information processing, like suicide, is a synchronous integration of cognitive functions that determine behaviour, motivation, and physiology (Rogers, et al., 2024). Moreover, stress and hopelessness appear to have a reciprocal relationship that can destabilize one another over time (Rogers, et al., 2024). However, Rogers, et al. (2024) also found that stress returns to a stable, equilibrium state faster than hopelessness, suicide rumination, and suicide intent. This further amplifies the complexity of human cognitive processes and reinforces that DST can explain such phenomena that are otherwise inexplicable.

In contrast to stability, flexibility involves the abandonment of past ways in favour of shifting responses to match new situations (Smith, 2009). Another fundamental component of human development is the role of executive function, a set of systems including cognitive flexibility, inhibitory control, and working memory (Perone, et al., 2021). As it turns out, executive control tasks such as the A-not-B task require flexibility to adapt to the changing environment in spite of a previous, stable experience (Smith, 2009). In such a goal-directed task, the capacity to assemble various components of attention, movement, language, and familiarity is highlighted by dynamic systems (Perone, et al., 2021). Smith (2009) expounds on this adaptive capacity by emphasizing that flexibility refers to the strength to surpass a threshold for past memories. As expected, DST denotes that alterations to the relative input strengths will alter the systematic response (Smith, 2009). Thus, Smith (2009) argues that flexibility and perseverance should rely on the dynamics of strength for a specific cue and its influence on the system.

Human cognition is restricted by most recent events at any given moment. However, at each moment, external sensory input acts to perturb these internal processes, infringing upon system dynamics (Smith, 2009). Importantly, the sensorimotor system is dynamically linked to motor memories and sensory events (Smith, 2009). In this way, motor plans, including moving a hand from its current location to a target location, are conceptualized to be flexible activations in the dynamic system (Smith, 2009). Similarly, memories of past hand movements also constitute motor plans, embodying a perseverative response (Smith, 2009). To act upon these cues, there must be a cognitive process that overcomes the hand’s stable state to change its position. Dynamic Systems Theory therefore emphasizes the tension between stability and flexibility that requires balance in order to maintain normal cognition.

Conclusion

Dynamic Systems Theory provides a novel take on a multitude of well-known theories by transforming them into a system of ever-evolving variables. As stress, development, and stability are nonlinear yet essential components of human cognitive function, a dynamic approach to conceptualizing these ideas reigns over stage diagrams. Stress, known for its chaotic spikes, lends itself to a differential system of chaos equations. Similarly, development described as a nonlinear map is much more reflective of real-life developmental patterns, where humans do not simply follow a discrete phase model. And lastly, the human struggle between stability and flexibility is not to be glossed over. Therefore, DST prevails in replicating the inconsistencies and eccentricities of the human cognitive system.

References

Benight, C. C., Shoji, K., & Delahanty, D. L. (2017). Self‐Regulation Shift Theory: A Dynamic Systems Approach to Traumatic Stress. *Journal of Traumatic Stress*, *30*(4), 333–342. https://doi.org/10.1002/jts.22208

Braun, U., Schäfer, A., Walter, H., Erk, S., Romanczuk-Seiferth, N., Haddad, L., Schweiger, J. I., Grimm, O., Heinz, A., Tost, H., Meyer-Lindenberg, A., & Bassett, D. S. (2015). Dynamic reconfiguration of frontal brain networks during executive cognition in humans. Proceedings of the National Academy of Sciences - PNAS, 112(37), 11678–11683. https://doi.org/10.1073/pnas.1422487112

Favela, L. H. (2020). Cognitive science as complexity science. Wiley Interdisciplinary Reviews. Cognitive Science, 11(4), e1525-n/a. https://doi.org/10.1002/wcs.1525

Favela, L. H. (2020). Dynamical systems theory in cognitive science and neuroscience. *Philosophy Compass*, *15*(8). https://doi.org/10.1111/phc3.12695

Keenan, E. K. (2010). Seeing the Forest and the Trees: Using Dynamic Systems Theory to Understand "Stress and Coping" and "Trauma and Resilience." *Journal of Human Behavior in the Social Environment*, *20*(8), 1038–1060. https://doi.org/10.1080/10911359.2010.494947

Knight, R. (2014). A Hundred Years of Latency: From Freudian Psychosexual Theory to Dynamic Systems Nonlinear Development in Middle Childhood. Journal of the American Psychoanalytic Association, 62(2), 203–235. https://doi.org/10.1177/0003065114531044

Li, N., Feng, L., Zhang, H., Cao, L., & Wang, X. (2024). Stress Prediction Based on Chaos

Theory and an Event-Behavior-Stress Triangle Model. *IEEE Transactions on Computational Social Systems*, *11*(5), 6056–6071. https://doi.org/10.1109/TCSS.2024.3386752

McClelland, James L., and Gautam Vallabha (2009). Dynamic Field Theory and Embodied Cognitive Dynamics. In Toward a Unified Theory of Development Connectionism and Dynamic System Theory Re-Consider (pp. 3-24). Oxford University PressNew York. https://doi.org/10.1093/acprof:oso/9780195300598.003.0005

Perone, S., Simmering, V. R., & Buss, A. T. (2021). A Dynamical Reconceptualization of Executive-Function Development. Perspectives on Psychological Science, 16(6), 1198–1208. https://doi.org/10.1177/1745691620966792

Rogers, M. L., Law, K. C., Lawrence, O. C., & Mandel, A. A. (2024). Perseveration on suicidal thoughts and images in daily life: An examination of the cognitive model of suicide through a dynamic systems lens. *Behaviour Research and Therapy*, *177*, 104524-. https://doi.org/10.1016/j.brat.2024.104524

Samuelson, L. K., Jenkins, G. W., & Spencer, J. P. (2015). Grounding Cognitive-Level Processes in Behavior: The View From Dynamic Systems Theory. Topics in Cognitive Science, 7(2), 191–205. https://doi.org/10.1111/tops.12129

Schöner, G. (2009). Development as Change of System Dynamics: Stability, Instability, and Emergence. In *Toward a Unified Theory of Development Connectionism and Dynamic System Theory Re-Consider*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195300598.003.0002

Smith, Linda B., Dynamic Field Theory and Embodied Cognitive Dynamics. In Toward a

Unified Theory of Development Connectionism and Dynamic System Theory Re-Consider (pp. 67-85). Oxford University PressNew York. https://doi.org/10.1093/acprof:oso/9780195300598.003.0005